Thermal Stresses in a Compound Bar

Problem

M5: A steel bar is encased in a copper sheath along its entire length, with the sheath securely bonded to the bar, preventing independent expansion. The cross-sectional area of the copper sheath is half that of the steel bar. Determine the stresses in both the steel and the copper when the compound bar is subjected to a temperature increase of 100°C.

Given Data

- Coefficient of linear expansion for steel: $\alpha_s = 12 \times 10^{-6}$ per $^{\circ}C$
- Coefficient of linear expansion for copper: $\alpha_c = 17 \times 10^{-6}$ per $^{\circ}C$
- Young's modulus of steel: $E_s = 206$ GN/m^2
- Young's modulus of copper: $E_c = 103$ GN/m^2
- Temperature increase: $\Delta T = 100^{\circ}C$
- Area ratio: $A_c = \frac{1}{2}A_s$

Solution

Since the bar is constrained, the free thermal expansion of steel and copper is prevented, resulting in thermal stresses. The free expansion strains would be:

$$\varepsilon_s = \alpha_s \Delta T, \quad \varepsilon_c = \alpha_c \Delta T$$

Substituting the values:

$$\varepsilon_s = (12 \times 10^{-6})(100) = 0.0012$$

$$\varepsilon_c = (17 \times 10^{-6})(100) = 0.0017$$

Let σ_s and σ_c be the stresses in steel and copper, respectively. Since the bar is constrained, the actual strain in both materials must be the same (ε):

$$\varepsilon = \varepsilon_s + \frac{\sigma_s}{E_s} = \varepsilon_c - \frac{\sigma_c}{E_c}$$

Since forces must balance **(1 mark)**:

$$\sigma_s A_s = \sigma_c A_c$$

Substituting $A_c = \frac{1}{2}A_s$:

$$\sigma_s = \frac{1}{2}\sigma_c$$

Solving for σ_s and σ_c, we substitute these relationships into the strain equation:

$$0.0012 + \frac{\sigma_s}{206 \times 10^9} = 0.0017 - \frac{\sigma_c}{103 \times 10^9}$$

Using $\sigma_s = \frac{1}{2}\sigma_c$:

$$0.0012 + \frac{\sigma_c}{2(206 \times 10^9)} = 0.0017 - \frac{\sigma_c}{103 \times 10^9}$$

Simplifying:

$$0.0012 + \frac{\sigma_c}{4.12 \times 10^{11}} = 0.0017 - \frac{\sigma_c}{1.03 \times 10^{11}}$$

Multiplying by 4.12×10^{11} to clear fractions:

$$(0.0012)(4.12 \times 10^{11}) + \sigma_c = (0.0017)(4.12 \times 10^{11}) - 4\sigma_c$$

Solving for σ_c **(2 marks)**:

$$4\sigma_c + \sigma_c = (0.0017 - 0.0012)(4.12 \times 10^{11})$$

$$5\sigma_c = (0.0005)(4.12 \times 10^{11})$$

$$\sigma_c = \frac{2.06 \times 10^8}{5} = 41.2 \text{ MPa (compressive)}$$

Using $\sigma_s = \frac{1}{2}\sigma_c$ **(2 marks)**:

$$\sigma_s = \frac{41.2}{2} = 20.6 \text{ MPa (tensile)}$$

Final Answers

- The stress in steel: **20.6 MPa (tensile)**
- The stress in copper: **41.2 MPa (compressive)**